

Lecture 15 – Network modules (3-spp.)

Today:

- Competition for explicit resources
- Trophic chain
- Intraguild predation

Concepts:

- Competitive exclusion principle
- Bottom-up and top-down effects
- Temporal niche partitioning (Armstrong-McGehee effects)

Method:

- Graphical analysis
 - Invasibility analysis
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Recall:

2-spp. LV competition with implicit resource

⇒ Intra > Inter-specific competition for coexistence

Same general insight from n -spp. community matrix analysis:

$$\lambda^2 - \underbrace{(A_{11} + A_{22})}_{\text{Trace}} \lambda + \underbrace{A_{11}A_{22} - A_{12}A_{21}}_{\text{Determinant}}$$

⇒ More zero off-diagonals (lower connectance), the more likely $\text{Det}(A) > 0 \Rightarrow$ more stable.

Exploitative Competition for Explicit Resource

Start with 2 consumer species competing for explicit resource.

Include self-limitation in resource

(Know that strong self-limitation on some species is needed for stable coexistence to be possible)

$$\begin{aligned}\frac{dO}{dt} &= \overbrace{e_{RO}a_{RO}RO}^{\text{linear intxn}} - \overbrace{m_oO}^{\text{exponential}} \\ \frac{dC}{dt} &= e_{RC}a_{RC}RC - m_cC \\ \frac{dR}{dt} &= rR \left(1 - \frac{R}{K}\right) - a_{RC}RC - a_{RO}RO\end{aligned}$$

Today's class is going to be 'Active learning' style

Groups of 4-5 people. (1 whiteboard per group.)

Use the methods we've learned in class to answer (referring to notes is okay)...

Questions for each module:

1. How many equilibria exist? (⇒ What are they symbolically?)
2. When can each species invade? (⇒ When are equilibria stable?)

In other words: **How can two consumers coexist on a single resource?**

Method: Graphical analysis of isoclines

Hint: Best to start analyzing $\frac{dO}{dt}$ and $\frac{dC}{dt}$ rather than $\frac{dR}{dt}$

.....
Let groups work for 20-30 min. before volunteer group explains to rest, or work through it together.
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Answer:

In order for a species to invade, its per capita growth rate must be positive: $\frac{1}{N} \frac{dN}{dt} > 0$

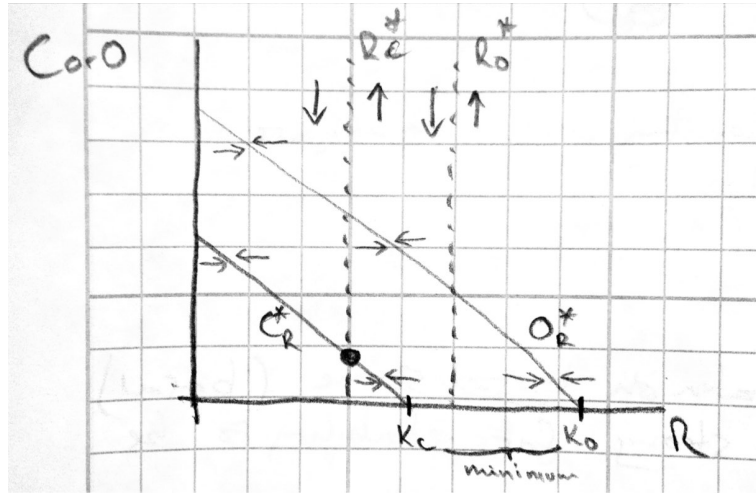
Predator isoclines: (Not really *isoclines*, but rather *isoplanes*.)

$$\begin{aligned} \frac{1}{O} \frac{dO}{dt} = e_{RO}a_{RO} - m_0 = 0 & \Rightarrow R_O^* = \frac{m_O}{e_{RO}a_{RO}} \\ \frac{1}{C} \frac{dC}{dt} = e_{RC}a_{RC} - m_C = 0 & \Rightarrow R_C^* = \frac{m_C}{e_{RC}a_{RC}} \end{aligned}$$

Prey isocline(s):

$$\begin{aligned} \frac{dR}{dt} = rR \left(1 - \frac{R}{K}\right) - a_{RC}RC - a_{RO}RO = 0 & = r - \frac{rR}{K} - a_{RC}C - a_{RO}O \\ \Rightarrow O_R^* = \frac{r - \frac{r}{K}R - a_{RC}C}{a_{RO}} \\ \Rightarrow C_R^* = \frac{r - \frac{r}{K}R - a_{RO}O}{a_{RC}} \end{aligned}$$

Note that O_R^* and C_R^* are linearly declining functions of both R and C or O .
Look at marginal view of phase portrait (i.e. 3D in 2D):



Conclusions:

1. K_C and K_O are minimum carrying capacities needed for coexistence of *either* (i.e. one or the other) consumer with R .
2. Whichever species has the lower R^* out-competes the other!
3. R^* is a metric for competitive superiority.
 C out-competes O if $R_C^* < R_O^*$...which means that $m_C < m_O$ and/or $e_{RC}a_{RC} > e_{RO}a_{RO}$.
4. Competitive exclusion occurs regardless of carrying capacity (K doesn't show up in either R_i^*).
As long as $K > \min R_i^*$, there will be only 1 consumer species.

But note assumption of linear species intxns. (Will get back to this.)

Trophic Chain

$$\begin{aligned}\frac{dO}{dt} &= e_{CO}a_{CO}CO - m_OO \\ \frac{dC}{dt} &= e_{RC}a_{RC}RC - a_{CO}CO - m_C C \\ \frac{dR}{dt} &= rR \left(1 - \frac{R}{K}\right) - a_{RC}RC\end{aligned}$$

Intuition from previous (exploitative competition) model:

Need some minimum amount of energy/productivity to support higher trophic levels.

Empirically, ~90% of energy is lost at each trophic transfer.

Questions:

1. How many equilibria and what are they (*qualitatively*)?
2. What level of productivity (K) is needed for each consumer to invade (*symbolically*)?

.....
① Trivial equilibrium $\Rightarrow R_{eq}^* = C_{eq}^* = O_{eq}^* = 0$

② Resource-only $\Rightarrow R_{eq}^* = K$

③ When can C invade R -only system?

In order to invade $\frac{1}{C} \frac{dC}{dt} > 0$.

Consumer isocline:

$$\frac{1}{C} \frac{dC}{dt} = e_{RC}a_{RC}R - \cancel{a_{CO}CO}^0 - m_C = 0 \quad \Rightarrow \quad R_C^* = \frac{m_C}{e_{RC}a_{RC}}$$

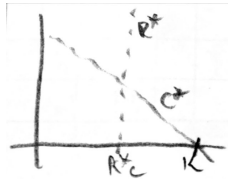
Resource isocline:

$$\frac{1}{R} \frac{dR}{dt} = r - \frac{rR}{K} - a_{RC}C = 0 \quad \Rightarrow \quad C_R^* = \frac{r - \frac{r}{K}R}{a_{RC}} = \frac{r}{a_{RC}} - \frac{r}{Ka_{RC}}R$$

Conclusion:

C can invade when $K > R_C^*$.

The R -only system becomes *unstable* when $K \geq R_C^*$.



④ O can't invade R -only system. When can O invade an RC -system at equilibrium?

RC equilibria:

$$R_{eq}^* = \frac{m_C}{e_{RC}a_{RC}}$$

$$C_{eq}^* = \frac{r - \frac{r}{K}R^*}{a_{RC}} = \frac{r - \frac{r}{K} \frac{m_C}{e_{RC}a_{RC}}}{a_{RC}} = \frac{r}{a_{RC}} - \frac{rm_C}{a_{RC}^2 e_{RC} K} = \frac{r(a_{RC}e_{RC}K - m_C)}{a_{RC}^2 e_{RC} K}$$

Invasion threshold C_O^* :

$$\frac{1}{O} \frac{dO}{dt} = 0 \quad \Rightarrow \quad C_O^* = \frac{m_O}{e_{CO}a_{CO}}$$

When does invasion threshold equal equilibrium (ie. $C_O^* = C_{eq}^*$)?

$$C^* = \frac{m_O}{e_{CO}a_{CO}} = \frac{r(a_{RC}e_{RC}K - m_C)}{a_{RC}^2e_{RC}K}$$

$$m_Oa_{RC}^2e_{RC}K = e_{CO}a_{CO}ra_{RC}e_{RC}K - e_{CO}a_{CO}rm_C$$

$$m_Oa_{RC} = e_{CO}a_{CO}r - \frac{e_{CO}a_{CO}rm_C}{a_{RC}e_{RC}K}$$

$$\frac{m_Oa_{RC}}{e_{CO}a_{CO}} = r - \frac{rm_C}{a_{RC}e_{RC}K}$$

$$\Rightarrow r - \frac{rm_C}{a_{RC}e_{RC}K} - \frac{m_Oa_{RC}}{e_{CO}a_{CO}} \geq 0$$

Invasion promoted by $\begin{cases} \text{Higher productivity (K)} \\ \text{Higher conversion efficiencies (} e_{RC} \text{ and } e_{RO} \text{)} \\ \text{Lower mortality rates (} m_C \text{ and } m_O \text{)} \end{cases}$

Note that conversion efficiencies and mortality rates contribute to ‘ecosystem efficiency’
Effects of r and a_{RC} depend on mortality rates.

④b) At what K can O invade?

Solve $C_O^* - C_{eq}^* = 0$ for K

$$m_Oa_{RC}^2e_{RC}K = e_{CO}a_{CO}ra_{RC}e_{RC}K - rm_Ce_{CO}a_{CO}$$

$$K(m_Oa_{RC}^2e_{RC} - e_{CO}a_{CO}ra_{RC}e_{RC}) = -rm_Ce_{CO}a_{CO}$$

$$K \geq \frac{-rm_Ce_{CO}a_{CO}}{(m_Oa_{RC}^2e_{RC} - e_{CO}a_{CO}ra_{RC}e_{RC})} = \frac{rm_Ce_{CO}a_{CO}}{(e_{CO}a_{CO}ra_{RC}e_{RC} - m_Oa_{RC}^2e_{RC})}$$

Repeat with Mathematica

Intraguild predation

Contains exploitative competition, trophic chain *and* apparent competition module!

$$\frac{dO}{dt} = e_{RO}a_{RO}RO + e_{CO}a_{CO}CO - m_OO$$

$$\frac{dC}{dt} = e_{RC}a_{RC}RC - a_{CO}CO - m_CC$$

$$\frac{dR}{dt} = rR \left(1 - \frac{R}{K}\right) - a_{RC}RC - a_{RO}RO$$

Questions:

1. How many equilibria and what are they (*qualitatively*)?
2. What level of productivity (K_i) allows each consumer to invade (*symbolically*)?

①②

Intuition from exploitative competition model:

Invasion thresholds in absence of other consumer:

$$\frac{1}{O} \frac{dO}{dt} = 0 \quad \Rightarrow \quad R_O^* = \frac{m_O}{e_{RO}a_{RO}}$$

$$\frac{1}{C} \frac{dC}{dt} = 0 \quad \Rightarrow \quad R_C^* = \frac{m_C}{e_{RC}a_{RC}}$$

If K is low enough, R -only is stable. (i.e. if $K < \min(R_i^*)$)

Consumer with lower R^* invades first. ③④

If both consumers are present:

$$\begin{aligned}\frac{1}{O} \frac{dO}{dt} = 0 & \Rightarrow R_O^* = \frac{m_O \boxed{-a_{CO}e_{CO}C}}{e_{RO}a_{RO}} \\ \frac{1}{C} \frac{dC}{dt} = 0 & \Rightarrow R_C^* = \frac{m_C \boxed{+a_{CO}O}}{e_{RC}a_{RC}}\end{aligned}$$

Therefore, C can never invade RO -equilibrium *unless...*

$$\frac{m_C}{e_{RC}a_{RC}} << \frac{m_O}{e_{RO}a_{RO}}$$

...which means that either $m_C << m_O$ and/or $e_{RC}a_{RC} >> e_{RO}a_{RO}$.

Conclusion: Coexistence requires that IGprey is be superior competitor for shared resource. Otherwise RO -system occurs.

⑤

When can O invade stable RC -system?

RC -system at equilibrium (exactly as for trophic chain):

$$R_{eq}^* = \frac{m_C}{a_{RC}e_{RC}} \quad C_{eq}^* = \frac{r(a_{RC}e_{RC}K - m_C)}{a_{RC}^2e_{RC}K}$$

Invasion threshold

$$\frac{1}{O} \frac{dO}{dt} = 0 \Rightarrow C_O^* = \dots \textbf{Mathematica}$$

Solve $C_O^* - C_{eq}^* = 0$ for K ...

$$K \geq \frac{a_{CO}e_{CO}m_C r}{a_{RC}(a_{RO}e_{RO}m_C - a_{RC}e_{RC}m_O + a_{CO}e_{CO}e_{RC}r)}$$

Show in Mathematica

⑥

When does C go extinct at high K ?

Solve $C_{eq}^* = 0$ from RCO -equilibrium for K :

$$K \geq \frac{a_{CO} \overbrace{m_O r}}{\underbrace{a_{RO}}(a_{RO}e_{RO}m_C - a_{RC}e_{RC}m_O + a_{CO}e_{CO} \underbrace{r})}$$

compare to solution in ⑤

Temporal niche partitioning

Conclusion from exploitative competition model: $n \geq 2$ consumers cannot coexist on 1 resource. But, among assumptions of this model was that interactions were linear.

Add Type II functional response to one of the two consumers.

$$\begin{aligned}\frac{dO}{dt} &= e_{RO} \frac{a_{RO}RO}{1 + a_{RO}h_{RO}R} - m_O O \\ \frac{dC}{dt} &= e_{RC}a_{RC}RC - m_C C \\ \frac{dR}{dt} &= rR \left(1 - \frac{R}{K}\right) - a_{RC}RC - \frac{a_{RO}RO}{1 + a_{RO}h_{RO}R} RO\end{aligned}$$

$\Rightarrow$ Mathematica Handling time induces positive density-dependence in prey, which induces cycles.

Cycles of RO -system induces cycles in C

C utilize resource when O is at low abundance and R is high.

Armstrong & McGehee 1980 also showed that $n \geq 4$ consumers can coexist on 4 resources.